

Abstract

This diploma project presents the development and evaluation of a multimodal emotion recognition system based on an attention bottleneck mechanism. The system addresses the CMU-MOSEI six-emotion multi-label classification task and uses three input modalities: text, audio, and visual information. The target emotion categories are happiness, sadness, anger, surprise, disgust, and fear.

The proposed architecture combines frozen BERT text representations, COVAREP acoustic features, and OpenFace facial action unit features. These modality-specific representations are projected into a shared hidden space and fused through 16 learnable bottleneck tokens. This design allows controlled cross-modal information exchange without full pairwise attention between all modality tokens.

The model was evaluated using macro-averaged F1 score as the main metric, with average weighted accuracy and per-class F1 used as additional measures. The proposed bottleneck fusion model outperformed the implemented late fusion and MulT-style baselines. It achieved an Avg. F1 of 0.4854 with the default threshold and 0.4942 after per-class threshold tuning. The largest improvements were observed for minority emotions, especially fear and surprise.

The project also includes a prototype FastAPI backend that processes raw video input, extracts text, audio, and visual features, and returns emotion probabilities through a REST API. The results show that attention bottleneck fusion can improve multimodal emotion recognition under class imbalance, while feature quality and preprocessing speed remain important limitations for future work.

Keywords: multimodal emotion recognition, CMU-MOSEI, attention bottleneck mechanism, BERT, COVAREP, OpenFace, macro-F1

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Chapter 1

Introduction

Emotion recognition has become an important research direction in artificial intelligence because modern digital communication increasingly relies on text, audio, and video. People express emotions not only through the meaning of words, but also through voice tone, facial expressions, and other non-verbal signals. As communication moves into online platforms, video conferencing, educational environments, and intelligent assistants, automatic understanding of human affect becomes more relevant for both research and practice.

In affective computing, emotion recognition aims to identify human emotional states from observable signals. Such systems may be useful in human–computer interaction, digital education, mental health monitoring, social media analysis, and assistive technologies. Kalateh et al. describe multimodal emotion recognition as a field that combines different sources of human expression in order to improve the reliability of emotion analysis [4]. This is especially important because human emotions are complex and cannot always be correctly interpreted from a single source of information.

Unimodal systems rely on only one modality, such as text, speech, or facial expression, which has clear limitations. Text may show semantic meaning, but it does not fully describe how the speaker said something. Audio can capture vocal cues such as pitch and rhythm, but may not contain enough semantic information. Visual features represent facial expressions and head movements, but they may be unreliable when video quality is low. For this reason, multimodal emotion recognition combines textual, acoustic, and visual information to obtain a more complete representation of human emotion [7].

The CMU-MOSEI dataset is one of the most widely used benchmarks for multimodal sentiment and emotion analysis. Zadeh et al. introduced it as a large-scale dataset collected from online videos, containing text, audio, and visual modalities [20]. CMU-MOSEI includes six emotion categories: happiness, sadness, anger, fear, disgust, and surprise. A major challenge is class imbalance, as happiness accounts for over 53% of samples, while fear and surprise are rare. Nguyen et al. emphasize that imbalanced datasets require special attention to minority classes and appropriate evaluation metrics [11]. For this reason, this project uses macro-F1 as the main metric, which gives equal importance to all emotion classes.

Another key challenge is multimodal fusion. Transformer-based models such as the Multimodal Transformer by Tsai et al. introduced cross-modal attention for unaligned sequences and demonstrated the importance of learning interactions between modalities [17]. However, full cross-modal attention can be computationally expensive. Attention bottleneck mechanisms address this limitation. Nagrani et al. proposed the Multimodal Bottleneck Transformer, where a small set of bottleneck tokens controls information exchange between modalities instead of allowing all tokens to attend to each other directly [10]. Later works adapted this idea to emotion recognition. He et al. proposed a domain-separated bottleneck attention fusion framework [2], and Nguyen et al. extended bottleneck transformers to address class imbalance through adaptive loss design [11].

Research Aim

The aim of this diploma project is to develop and evaluate a multimodal emotion recognition system based on an attention bottleneck mechanism for six-emotion classification using textual, acoustic, and visual modalities from the CMU-MOSEI dataset.

Research Objectives

To achieve this aim, the following objectives are defined:

- to study existing approaches to multimodal emotion recognition and attention-based fusion methods;
- to analyze the CMU-MOSEI dataset and prepare multimodal features for training;

- to design and implement a multimodal architecture using an attention bottleneck mechanism;
- to train the model under class imbalance conditions using weighted loss and regularization techniques;
- to evaluate the system using macro-F1, weighted accuracy, and per-class metrics;
- to compare the proposed bottleneck fusion approach with a simpler fusion baseline.

The research is guided by several main questions. First, how can textual, acoustic, and visual features be effectively combined for emotion recognition on CMU-MOSEI? Second, whether an attention bottleneck mechanism can improve multimodal fusion compared to a simpler late fusion baseline. Third, how class imbalance affects recognition of rare emotions such as fear and surprise. Fourth, which evaluation metrics provide the most meaningful assessment of model performance in an imbalanced multimodal setting.

The scope of this project is limited to prototype-level development and experimental evaluation. The system uses precomputed multimodal features instead of raw audio and video processing, which allows the research to focus on multimodal fusion rather than low-level feature extraction. The system is not intended as a production-ready real-time application.

The rest of this report is organized as follows. Chapter 2 reviews existing research on multimodal emotion recognition, CMU-MOSEI, feature extraction, fusion methods, and bottleneck attention mechanisms. Chapter 3 describes the research methodology, dataset preparation, preprocessing, evaluation metrics, and experimental setup. Chapter 4 presents the system design and implementation. Chapter 5 reports the experimental results. Chapter 6 discusses the findings, limitations, and future work. Chapter 7 summarizes the main conclusions.

Chapter 2

Literature Review

2.1 Emotion Recognition in Human Communication

Emotions are a fundamental part of how people communicate in nowadays world. When someone speaks, they do not only use words to express their thoughts — they rely on the tone of their voice, their facial expressions, and their body language to convey how they feel. For a long time, computers were simply able to understand the content of speech, not the way in which it was spoken or the speaker’s facial expressions. The field of emotion recognition research aims to alter this by allowing machines to recognize and analyse human emotional states from observable information.

Kalateh et al. conducted a comprehensive systematic review of multimodal emotion recognition and described it as a field that ”combines different types of human signals in order to improve the reliability of emotion analysis” [4]. Their review covers a wide range of applications where emotion recognition is already being used or studied: human-computer interaction, healthcare, education, social robotics, gaming, and intelligent assistants. What stands out in their findings is that none of these domains can rely on a single signal alone — in healthcare, for example, a patient might say they feel fine while their voice and face suggest otherwise. This gap between what is said and what is actually felt is exactly why automatic emotion recognition is both difficult and important.

The field did not start with the sophisticated systems we see today. Koromilas and Giannakopoulos traced the development of multimodal emotion recognition from its earliest stages, showing that the first systems relied almost entirely on handcrafted features and classical machine learning classifiers like Support Vector Machines [7]. These

early methods worked decently in controlled laboratory environments, but they had trouble with real-world data, including noisy speech, partially visible faces, and subtle or ambiguous emotional expressions. This was fundamentally changed as the transition toward deep learning; recurrent neural networks (RNN), convolutional networks, and eventually transformer-based models. All this made it possible to learn complex patterns straight from data without needing to manually describe what an emotional signal looks like.

One of the core reasons multimodal approaches outperform unimodal ones is explained clearly by Kalateh et al., who note that using several modalities together allows the system to become more robust, because one modality can compensate for the weakness of another [4]. A classic example is sarcasm — a sentence that sounds positive in text might carry a completely different meaning when combined with a flat or negative vocal tone. Similarly, a spoken phrase might be unclear without seeing the speaker’s face. Neither text, audio or video alone tells the full story. Koromilas and Giannakopoulos further support this by pointing out that “the two key challenges in multimodal learning are learning complex interactions within and across modalities, and building models that are robust when one modality is missing or noisy at test time” [7].

This project sits directly within this context. The system developed there by using an attention bottleneck fusion technique to deal with three modalities: text, audio, and visual. The purpose is to learn how various signals interact and what occurs when the model must handle uncommon emotions that appear infrequently in training data. This goal goes far beyond merely creating a system that accurately identifies the most prevalent emotion. The challenges highlighted by Kalateh et al. directly shaped the design of this project. Specifically, modality heterogeneity, class imbalance, and the necessity for effective cross-modal interaction motivated the architectural and training choices detailed in subsequent chapters, a consensus further supported by Koromilas and Giannakopoulos[7].

2.2 The CMU-MOSEI Dataset

Datasets are important in multimodal emotion recognition because they define what kind of emotional signals a model can learn from. For this task, the dataset should include text, audio, and visual information, since emotion is usually expressed through language, voice, and facial behavior together. Zadeh et al. introduced CMU-MOSEI as

a large-scale collection for multimodal language analysis collected from online videos [20]. The authors describe it as a dataset based on natural monologue-style recordings, where speakers express opinions in real-world conditions. This makes it useful for realistic emotion recognition research, but also more difficult than controlled laboratory datasets.

In the original paper, Zadeh et al. explain that the dataset contains three main modalities: language, acoustic signals, and visual signals [20]. Each modality carries a different type of affective information. Text provides the semantic meaning of an utterance. Audio reflects how the speaker says it through prosody, rhythm, and vocal intensity. Visual information describes facial behavior and other non-verbal cues. This multimodal structure makes CMU-MOSEI suitable for studying cross-modal learning, because the model has to combine different sources of information instead of relying on only one channel.

Main study collection is also important because it supports both sentiment analysis and emotion recognition. Sentiment analysis usually focuses on polarity, such as positive or negative meaning, while emotion recognition is more detailed and focuses on separate emotion categories. Mora Melanchthon describes CMU-MOSEI emotion recognition as a categorical task where each emotion can be treated as present or absent [9]. This means that the task is not the same as simple single-label classification. One utterance may contain several emotional cues, and each emotion must be considered separately.

A major difficulty of CMU-MOSEI is the uneven distribution of emotion labels. Mora Melanchthon notes that happiness is the most frequent emotion in the dataset, while fear and surprise are among the rarest categories [9]. This imbalance affects how models learn. A model may recognize frequent emotions more easily, but fail on rare classes. Because of this, overall accuracy can be misleading. A system may look effective if it predicts common emotions correctly, even when it performs poorly on minority emotions.

Mora Melanchthon also shows that feature-level performance on dataset differs across modalities [9]. Text, acoustic, and visual features do not contribute equally to every emotion, and feature selection or transformation can affect model performance. This shows that multimodal emotion recognition depends not only on the fusion method, but also on how well each modality is represented before fusion. Overall, Zadeh et al. provide the foundation for CMU-MOSEI and its multimodal structure, while Mora Melanchthon highlights its emotion recognition setting, class imbalance, and modality-specific feature challenges [9, 20].

2.3 Feature Extraction for Text, Audio, and Video Modalities

Before a multimodal system can start learning, raw data must be converted into numerical representations that can be processed by a neural network. A video file itself is only a combination of visual frames and sound signals, and these signals need to be transformed into structured feature representations. This stage is known as feature extraction. In this project, three separate feature pipelines are used: one for text, one for audio, and one for visual data. Each pipeline is based on established tools and methods used in multimodal emotion recognition literature.

For the text modality, recent studies often use pre-trained language models instead of simple static word embeddings. Makhmudov et al. explain that using a pre-trained BERT-based model is useful because it has already learned rich linguistic representations from a large text corpus [8]. BERT, which stands for Bidirectional Encoder Representations from Transformers, processes text bidirectionally and therefore captures how each word is related to its surrounding context. This is important for emotion recognition because the emotional meaning of a word or phrase often depends on the full sentence. For example, the word “fine” may express a neutral meaning in one context but may also carry negative or sarcastic meaning in another context. In this project, BERT-base-uncased is used as a frozen feature extractor, and the final text representation is obtained by averaging the outputs of the last four hidden layers. A similar use of BERT-based textual representation is also found in video emotion recognition research by Xia et al. [18].

For the audio modality, this project uses COVAREP acoustic features. COVAREP is a collaborative voice analysis toolkit that extracts low-level acoustic descriptors from speech signals. It produces a 74-dimensional feature vector for each time frame and captures properties such as fundamental frequency, Mel-frequency cepstral coefficients, spectral envelope, glottal source parameters, and voiced–unvoiced segmentation indicators. These acoustic descriptors represent how a person speaks, including pitch, rhythm, energy, and voice quality. Such characteristics are important for emotion recognition because emotional states are often reflected in vocal behavior. For example, fear may be associated with changes in pitch and energy, while sadness may be expressed through slower and lower-energy speech. Mora Melanchthon discusses the use of acoustic features in CMU-MOSEI-based emotion recognition and shows that such features provide a compact and interpretable representation of emotional speech signals [9]. COVAREP-based acoustic

features are also commonly used in multimodal transformer-based studies on CMU-MOSEI [17].

For the visual modality, this project uses facial behavior features extracted with OpenFace. OpenFace is a toolkit that detects facial landmarks, head pose, eye gaze, and facial action units from video frames [9]. Facial action units describe the movement of specific facial muscle groups. For example, some action units correspond to cheek movement, lip corner movement, or brow movement. In this project, the visual representation consists of 35 values per frame, including action unit intensity scores and binary presence indicators. These features are not raw pixel values; they are compact numerical descriptors of facial behavior at each moment. This makes them easier to process and more interpretable than raw image frames, especially in datasets such as CMU-MOSEI where video quality and recording conditions may vary.

The use of handcrafted features such as COVAREP and OpenFace is a deliberate trade-off in this project. More recent models, such as the MER-SEM-MBT model proposed by Xia et al., use stronger pretrained neural encoders, including VGGish for audio and ResNet18 for visual features [18]. These encoders can produce richer representations and may achieve stronger benchmark results. However, they also require more computational resources and are less interpretable than compact handcrafted descriptors. Therefore, this project uses standard CMU-MOSEI feature representations in order to keep the system computationally feasible and comparable with widely used multimodal baselines such as MulT [17].

2.4 Multimodal Fusion Approach

When features are extracted from each modality, the next question is how to combine them. In general, this step is called multimodal fusion, and it plays a crucial role in the design of multimodal system. The modality fusion strategy determines whether the model can effectively capture cross-modal relationships between text, audio and video. Over the years, researchers have proposed many different fusion strategies, varying from straightforward to complex architectures. Therefore, understanding their trade-offs is very important for the frameworks.

The simplest strategies are early fusion and late fusion. In early fusion, features from all input streams are concatenated into a single vector before being passed to the model.

This is easy to implement, but it assumes that all modalities are equally important at every stage of processing. That is why one modality may be noisier or weaker than the others. In late fusion, each modality is processed by a completely separate model, and only the final predictions are combined. It allows each modality to be processed independently, but it reduces the capacity to learn fine-grained cross-modal feature interactions. As Nagrani et al. describe, late fusion is a useful baseline. However, full early fusion, where every token can attend to every other token across modalities, is computationally expensive and not always more accurate. This happens because much of the cross-modal attention is spent on redundant or irrelevant information [10].

The approach that changed the direction of the field for multimodal language analysis was the Multimodal Transformer, known as MulT, introduced by Tsai et al. [17]. Instead of simply concatenating features or averaging predictions, MulT introduced directional cross-modal attention, where one modality attends to another. For example, the language modality can attend to the audio sequence, asking at each word: which audio frames are most relevant to what is being said here? This property is particularly useful for CMU-MOSEI, where the modalities are not perfectly aligned in time. For instance, a facial expression may appear a fraction of a second before the corresponding word is spoken. MulT handles this naturally because cross-modal attention does not require alignment; it learns the correspondence between sequences directly from data. Tsai et al. showed that cross-modal attention consistently outperforms early and late fusion approaches on CMU-MOSEI. As a result, MulT has become a widely used benchmark in later studies.

Nevertheless, MulT has a limitation. When three modalities are used with bidirectional attention between every pair, the number of cross-modal attention operations increases rapidly. This leads to a situation where a considerable amount of attention is spent on information that is not always relevant for the final emotion prediction. Jia et al. also note that the audio modality often contributes less than text or vision in many multimodal models. They argue that more efficient fusion strategies can help balance the contribution of each modality without significantly reducing overall performance [3]. This observation is relevant to this project, where COVAREP audio features are already weaker than BERT text representations by nature, and the fusion mechanism needs to handle this imbalance thoughtfully.

Recent work has focused on combining the advantages of cross-modal attention with

more controlled information flow. Xia et al. introduced a model that adds a semantic enhancement step before fusion. It uses the text modality to guide how audio and visual representations are refined [18]. Their method is based on the assumption that text provides the most reliable high-level semantic information. By making audio and visual representations more text-aware before fusion, the model can learn better joint representations. This type of modality-guided fusion represents a more structured way of modeling cross-modal interactions. Instead of allowing all modalities to influence each other equally, some models explicitly assign a guiding role to the most informative modality.

All of these approaches share a common limitation. They allow full pairwise attention between all tokens across modalities, which scales quadratically with sequence length. As a result, the model exchanges a large amount of redundant information. The next section introduces a fundamentally different idea that this project builds on directly. The attention bottleneck restricts cross-modal information flow by passing it through a small set of shared latent tokens.

2.5 Attention Bottleneck Mechanism

More controlled way for modality of exchange information demonstrated by attention bottleneck mechanism. In emotion recognition, this is especially useful because text, audio, and visual signals do not contribute equally at every moment. Some words may be emotionally neutral, some audio frames may contain noise, and some facial movements may not express a clear body language. Full cross-modal attention can capture rich interactions, but it may also become computationally heavy and pass redundant information between modalities. Nagrani et al. address this problem through the Multimodal Bottleneck Transformer, where a small group of fusion tokens works as an intermediate communication channel between modality-specific streams [10]. Similar concerns are also discussed by He et al., who use bottleneck attention in a domain-separated fusion framework to control interaction between shared and modality-specific representations [2]. Nguyen et al. further promote this direction by extending bottleneck-based fusion to multimodal emotion recognition under class imbalance conditions [11]. Together, these studies show that bottleneck attention is not only a way to reduce computation. They act as a method for making cross-modal communication more selective and meaningful.

Nagrani et al. introduced this idea in the Multimodal Bottleneck Transformer (MBT) [10]. Instead of allowing focused dedication between all modality tokens, MBT uses a small number of fusion bottleneck tokens as an intermediate communication channel. Each modality sends information to these tokens, and the bottleneck tokens then share the compressed information back to the modality-specific streams. Their study shows that bottleneck-based fusion can reduce computational cost while still preserving important cross-modal interaction. This makes the mechanism useful for multimodal tasks where full pairwise attention may be unnecessary or inefficient.

The main strength of bottleneck tokens is their selectivity. Since the number of bottleneck tokens is limited, the model cannot simply transfer all information from one modality to another. It has to compress and prioritize the most important signals. Because emotional evidence is often localized to brief vocal or facial expressions, bottleneck attention provides a dual benefit: reducing computational costs and guiding the network toward sparse, meaningful multimodal features [10].

Later studies adapted bottleneck-based fusion more directly to multimodal emotion recognition. He et al. proposed the Domain-Separated Bottleneck Attention Fusion framework, where modality representations are separated into private and invariant domains [2]. The private domain preserves modality-specific information, while the invariant domain captures shared emotional information across modalities. This is useful because text, audio, and visual signals do not always express emotion in the same way. By applying bottleneck attention between these separated representations, the model can exchange information in a more structured way instead of simply mixing all features together [2].

Nguyen et al. also extended the bottleneck transformer approach in the Extended Multimodal Bottleneck Transformer (XMBT) [11]. Their work is especially relevant to CMU-MOSEI emotion recognition because it considers both multimodal fusion and class imbalance. XMBT uses modality-specific branches together with shared bottleneck tokens and adds auxiliary learning signals for individual modalities. The authors also introduce Dynamic Restrained Adaptive loss to enhance learning for underrepresented emotions. This showcase that bottleneck fusion integrates with targeted training strategies to mitigate modality dominance and class imbalance [11].

Overall, research points to computational and representational benefits of attention bottleneck approaches. They recommend the model to focus on clear, informative

cross-modal signals and prevent unneeded attention exchanges. Nevertheless, bottleneck attention by itself is not a whole solution. The training strategy, class imbalance management, and the quality of modality-specific elements continue to determine its efficacy. Bottleneck fusion is therefore combined with domain separation, secondary losses, and adaptive loss functions in order in modern studies [2, 11]. These findings provide the theoretical foundation for this project’s primary fusion approach that acts as an attention bottleneck mechanism.

2.6 Gaps in Existing Research and Project Justification

The reviewed literature shows that multimodal emotion recognition has improved significantly, but several problems remain open. One of the main difficulties is the recognition of rare emotions in CMU-MOSEI. Emotions such as fear and surprise appear much less often than happiness, so models can become biased toward frequent categories. Nguyen et al. discuss this problem and propose adaptive loss design to improve learning for underrepresented emotions [11]. He et al. also show that minority emotions are harder to classify because of class imbalance and overlapping emotional patterns [2]. For this reason, this project does not rely only on general accuracy. It considers weighted loss, threshold tuning, macro-F1, and per-class F1 analysis.

Another important gap concerns how different modalities are represented before fusion. Text, audio, and visual features have different structures and dimensions, so they are incapable to fuse directly without transformation. Encoders are needed to convert each modality into a numerical representation that the model can process, and to map all modalities into a shared hidden dimension. Some recent studies use more complex feature extraction pipelines at this stage. For example, Xia et al. use VGGish for audio and ResNet18 for visual information to obtain richer representations [18]. However, such approaches require more computational resources. This project utilizes standard CMU-MOSEI features. Specifically, modality-specific encoders project BERT-based text, COVAREP acoustic, and OpenFace-style visual features into a unified representation space prior to bottleneck fusion.

A further limitation is that many studies make it difficult to understand which part of the model actually improves performance. In some papers, several changes are added

at the same time: stronger feature extractors, new fusion blocks, auxiliary losses, or adaptive training strategies. As a result, it is not always clear whether the improvement comes from the fusion mechanism or from another component. Xia et al. show the value of semantic enhancement, while Nguyen et al. combine bottleneck fusion with auxiliary losses and adaptive loss design [11, 18]. This project addresses the same issue by comparing model variants under one experimental setting and analyzing the effect of different design choices.

Furthermore, a practical gap persists between benchmark models and deployable, production-ready systems. While literature extensively reports performance metrics on curated datasets, discussions regarding the integration of these models into operational applications remain limited. Kalateh et al. note that computational complexity, real-time performance, and deployment remain important challenges in multimodal emotion recognition [4]. This is relevant because an emotion recognition model should not only work in an isolated experiment, but also be understandable and extendable as part of a software system. Consequently, this project evaluates both model training and the deployment of the trained architecture within a prototype system.

In conclusion, these gaps justify the direction of this work. The project focuses on attention bottleneck fusion because it offers a controlled way to exchange information between text, audio, and visual modalities. Furthermore, this study addresses class imbalance, modality representation, and systematic evaluation. The objective extends beyond comparative score analysis to evaluate whether a compact, bottleneck-based fusion strategy can effectively support a realistic multimodal emotion recognition system utilizing standard CMU-MOSEI features.

2.7 Comparative Analysis of Related Works

Table 2.1 summarises the main related works reviewed in this chapter. The comparison focuses on the model type, used modalities, dataset, reported Avg. F1 where available, and the design idea that makes each study relevant to this project.

Table 2.1: Comparative analysis of related works in multimodal emotion recognition.

Study	Model Type	Modality	Dataset	Avg. F1	Key Notes
Tsai et al. (2019) [17]	Multimodal Transformer (MulT)	Text + Audio + Vision	CMU-MOSEI	0.475	Uses directional cross-modal attention for unaligned sequences. This work is important as a strong reference model for CMU-MOSEI.
Nagrani et al. (2021) [10]	Multimodal Bottleneck Transformer (MBT)	Audio + Vision	AudioSet, VGGSound	–	Introduces bottleneck tokens for multimodal fusion. The idea reduces full pairwise attention by passing information through shared latent tokens.
Xia et al. (2022) [18]	MER-SEM-MBT	Text + Audio + Vision	CMU-MOSEI	0.509	Combines semantic enhancement with bottleneck fusion. The model uses BERT, VGGish, and ResNet18, which gives stronger audio and visual representations.
He et al. (2024) [2]	Domain-Separated Bottleneck Attention	Text + Audio + Vision	CMU-MOSEI	–	Separates private and invariant modality representations. The study also supports the use of bottleneck attention for structured cross-modal exchange.
Nguyen et al. (2025) [11]	XMBT + DRA loss	Text + Audio + Vision	CMU-MOSEI	0.496	Extends bottleneck fusion to multimodal emotion recognition and adds adaptive loss for class imbalance. The work is especially relevant for rare emotions such as fear and surprise.
Sanjyal (2026) [14]	Causal-Diffusion Bridge	Text + Audio + Vision	CMU-MOSEI subset	0.214	Uses compact acoustic and visual features with a smaller aligned subset. The result shows that feature quality and data volume can strongly affect final scores.
Proposed work	BERT + Conv1D + Bottleneck Attention	Text + Audio + Vision	CMU-MOSEI	0.485	Uses frozen BERT, COVAREP, OpenFace, 16 bottleneck tokens, auxiliary losses, and modality dropout. The model focuses on compact feature representations and rare emotion recognition.

Chapter 3

Methodology

3.1 Research Design

This work is applied in nature because it focuses on developing and evaluating a functional multimodal emotion recognition system. It does not attempt to propose a new theory of human emotion. Instead, it addresses a practical machine learning task: combining text, audio, and visual information to classify six emotion categories in an imbalanced dataset. The main outcome is an operational neural network and a supporting processing pipeline that can transform multimodal inputs into emotion predictions.

The study follows a design-oriented research logic. The process begins with problem analysis through the literature review, where existing approaches, limitations, and evaluation practices are examined. Based on this analysis, the architecture is designed and then implemented. After that, the baseline and proposed models are trained under the same experimental conditions. The final stage is evaluation using quantitative metrics. This sequence is suitable for deep learning research, where architectures are commonly tested through controlled experiments on established datasets. Koromilas and Giannakopoulos also describe deep multimodal emotion recognition as a field where model development and experimental comparison are central parts of the research process [7].

The work has two connected directions. The first direction is experimental and focuses on model performance. A late fusion baseline and an attention bottleneck fusion model are trained on CMU-MOSEI and compared using the same validation and test protocol. This part answers the main research question about whether bottleneck-based fusion

improves recognition quality. The second direction is practical and concerns prototype integration. The trained model is prepared for use inside a processing pipeline that can later be connected to the MultiMOOD web interface. In this way, the study is not limited to a notebook experiment, but also considers how the developed solution could be used in an applied software setting.

The evaluation is quantitative. Interviews, surveys, and subjective user assessment are not included because the research questions focus on measurable model performance. The evidence is obtained from macro-F1, weighted accuracy, and per-class F1 scores computed on validation and test sets. These metrics make it possible to compare different variants and examine whether the proposed architecture improves recognition, especially for underrepresented emotion categories.

3.2 Quantitative Experimental Approach to Emotion Recognition

This study uses a quantitative experimental approach because its main objective is to measure the performance of machine learning models. In emotion recognition research, model quality is usually evaluated through numerical metrics computed on validation and test data. In this work, the main measures are macro-averaged F1 score, average weighted accuracy, and per-class F1 scores for the six emotion categories. These values allow the proposed bottleneck-based architecture to be compared with the late fusion baseline under identical conditions. Ramyasree et al. use a similar evaluation logic in their multimodal bottleneck transformer study, where accuracy and F1-score are reported as the main indicators of performance [13].

The quantitative approach is also important because CMU-MOSEI is imbalanced. A general accuracy score may not show whether rare emotions are recognized well. For this reason, macro-F1 and per-class F1 are used to examine results across all categories, not only the dominant ones. This is especially relevant for difficult classes such as fear and surprise.

The same logic is applied to the ablation study. Several configurations are tested under one controlled setup. Each configuration changes a specific part of the architecture or training strategy, such as BERT freezing, the number of averaged BERT layers, auxiliary losses, modality dropout, or threshold tuning. Macro-F1, weighted accuracy, and per-class

F1 are then compared to determine the effect of each design choice. This makes the analysis more systematic and helps identify which components contribute most to the final result.

3.3 Justification of the Experimental Evaluation Approach

The experimental evaluation approach was selected because it matches the purpose of this work. The central question is whether the attention bottleneck mechanism improves multimodal emotion recognition compared with a simpler fusion baseline. This type of question is best answered through controlled experiments, where both models are trained and tested on the same data under the same conditions. Interviews or subjective observations would not provide the main evidence in this case, because model quality must be assessed through measurable results on unseen samples.

Using a widely accepted dataset and standard metrics also makes the findings easier to compare with previous studies. Koromilas and Giannakopoulos describe deep multimodal emotion recognition as a field where architectures are often evaluated through experimental comparison and numerical results [7]. CMU-MOSEI is suitable for this purpose because it is widely used in multimodal emotion and sentiment recognition research. If a different dataset or unusual metrics were used, it would be harder to place the results in the context of existing work.

The selected metrics are connected to the properties of CMU-MOSEI. Since the data is imbalanced, plain accuracy may hide weak performance on rare emotions. Mora Melanchthon shows that macro-F1 is useful for CMU-MOSEI because it reflects recognition quality across separate categories more clearly than general accuracy [9]. Therefore, macro-F1 is used as the primary metric, while weighted accuracy is reported as a secondary measure. This strategy evaluates both overall behaviour and recognition of underrepresented classes.

3.4 Dataset and Computational Instruments

The main data source for this work is CMU-MOSEI. It provides the training, validation, and test samples used in the experimental part of the study. Zadeh et al. introduced

CMU-MOSEI as a large-scale dataset for multimodal language analysis collected from online videos [20]. It contains more than 23,000 annotated video segments and includes text, audio, and visual information. This structure is suitable for the present task because the system is designed to recognize emotions from three complementary input streams.

CMU-MOSEI already contains structured multimodal features and emotion labels. Each sample can be represented through textual, acoustic, and visual streams. This allows modality-specific encoders to process each source separately before fusion. Such a format is important for the proposed bottleneck module, where different inputs are first projected into a shared representation space and then combined. Related studies also use acoustic and visual descriptors such as COVAREP and OpenFace-style features, which makes the dataset appropriate for comparison with previous work [9, 14].

The computational instruments include the software libraries and hardware environment used for preprocessing, training, evaluation, and prototype integration. PyTorch is used as the main deep learning framework for implementing both the baseline and proposed architectures. The HuggingFace Transformers library provides BERT-based text representations. Scikit-learn is used to compute macro-F1, weighted accuracy, and per-class F1. All training experiments are conducted on the Kaggle platform with a Tesla T4 GPU, which is sufficient for the selected model size and batch configuration.

Additional tools support the deployment-oriented preprocessing pipeline. The openS-MILE toolkit extracts acoustic descriptors, Py-Feat detects facial action units, and faster-whisper transcribes speech. These tools allow the system to move beyond prepared dataset features and process raw video input in a prototype setting. Thus, the computational setup supports both experimental evaluation and practical implementation.

Figure 3.1 shows the tensor shape produced by each modality after preprocessing and projection. These shapes remain fixed across the experiments and define the input format expected by the bottleneck fusion model.

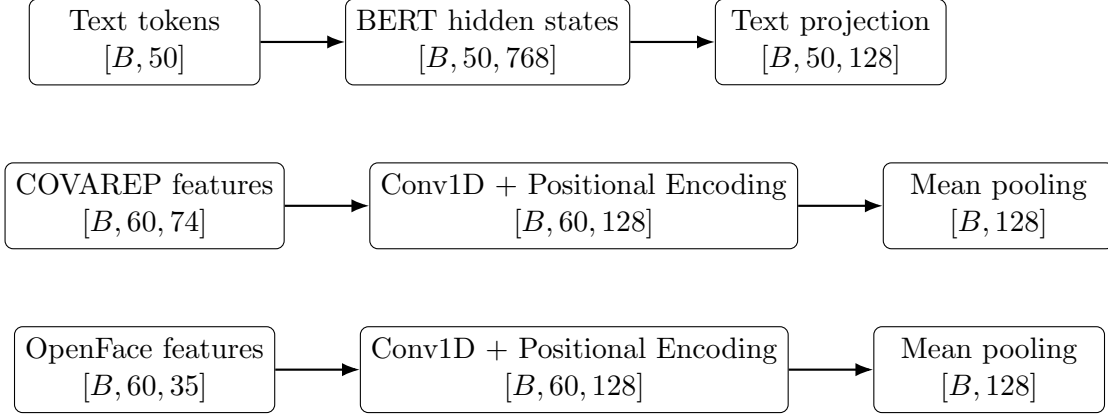


Figure 3.1: Feature tensor shapes for text, audio, and visual modalities.

3.5 Data Sampling and Class Imbalance in CMU-MOSEI

The experiments use the train, validation, and test split provided with CMU-MOSEI. A fixed split keeps the setting consistent and makes the results easier to compare with related studies. Before training, utterances with all-zero emotion labels are removed. Since this work focuses on six positive emotion categories, samples without any positive label are excluded from the supervised classification set. After filtering, the dataset contains 13,934 training samples, 1,569 validation samples, and 3,957 test samples. Nguyen et al. use a similar filtering strategy in their Extended Multimodal Bottleneck Transformer study on CMU-MOSEI [11].

The main sampling issue is class imbalance. The six emotion categories are not represented equally. Happiness is the dominant class and appears in approximately 62.7% of the training samples, while fear is the rarest category with only 9.6%. Surprise and disgust are also less frequent. This distribution affects learning because the classifier may recognize common emotions more easily and ignore minority classes. As a result, a model can show acceptable overall performance while still producing weak results for underrepresented categories. This is why macro-F1 and per-class F1 are important in this work.

Figure 3.2 shows the positive ratio for each emotion in the training set. The imbalance shown in the figure motivates several methodological decisions, including class-specific positive weights in the loss function and threshold tuning on the validation set. These

choices are introduced later in this chapter and described in more detail in the system design and training sections.

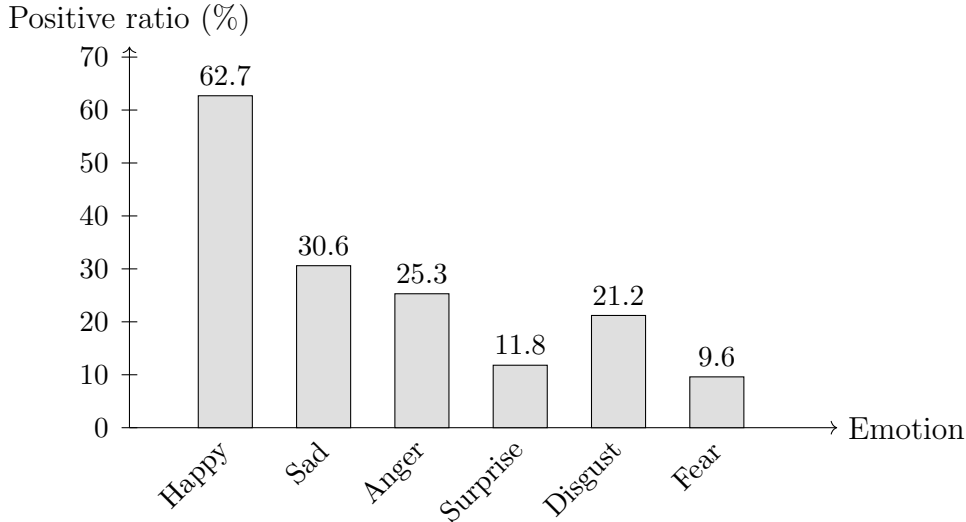


Figure 3.2: Class imbalance in the filtered CMU-MOSEI six-emotion dataset.

3.6 Multimodal Feature Preprocessing Pipeline

The preprocessing pipeline converts CMU-MOSEI multimodal features into a fixed tensor format required by the neural network. The original dataset introduced by Zadeh et al. contains text, audio, visual information, and emotion annotations collected from online video segments [20]. In this work, preprocessing does not create a new dataset from participants. It prepares the existing features for training, validation, testing, and later prototype inference.

For the text stream, the available word-level information is converted into sentence-level input and processed with a frozen BERT-base-uncased model. The final representation is obtained by averaging the last four hidden layers of BERT. This produces a tensor of shape $[B, 50, 768]$, where B is the batch size and 50 is the maximum sequence length. Before fusion, a linear projection maps this representation to the shared hidden dimension $[B, 50, 128]$.

For the audio stream, COVAREP descriptors are used. Each sequence contains 74-dimensional vectors over a fixed length of 60 time steps, producing a tensor of shape $[B, 60, 74]$. These values represent speech-related properties such as pitch, spectral

characteristics, and voice quality. For the visual stream, OpenFace-style facial features are used. Each sequence contains 35-dimensional facial action unit descriptors over 60 frames, producing a tensor of shape $[B, 60, 35]$. Audio and visual sequences are padded when they are shorter than the required length, and all tensors are converted to 32-bit floating point format.

After preprocessing, the prepared samples are stored in a pickle file for efficient loading. Each item contains text, audio, and visual tensors, a six-dimensional binary emotion label vector, and padding masks for audio and visual sequences. These masks indicate which frames are real observations and which are padding. This storage format allows batches to be loaded consistently across all experiments.

The same general input structure is considered for the prototype pipeline. For new video input, the system can extract audio with `ffmpeg`, compute acoustic descriptors with `opensmile`, detect facial action units with `py-feat`, and transcribe speech using `faster-whisper`. The recognized text is then tokenized and processed through BERT. This deployment-oriented flow is designed so that inference data follows the same general format as the training samples.

Table 3.1: CMU-MOSEI dataset statistics before preprocessing (raw HDF5 format).

Property	Value
Format	HDF5 (.h5)
Total segments	23,453
Splits	train / valid / test (fixed)
<i>Text modality</i>	
Field name	words
Representation	Raw byte strings (word tokens)
Sequence length	Variable (up to 50 words)
Dimension per token	— (not embedded)
<i>Audio modality</i>	
Field name	COVAREP
Representation	Float64 acoustic descriptors
Sequence length	Variable (up to 500+ frames)
Dimension per frame	74
<i>Visual modality</i>	
Field name	OpenFace_2
Representation	Float64 facial action units
Sequence length	Variable
Dimension per frame	713 (full feature set)
<i>Labels</i>	
Field name	All Labels
Format	Float scores per emotion
Emotions	happy, sad, anger, surprise, disgust, fear
Samples with all-zero labels	3,013 (excluded)
<i>Emotion distribution (train split, raw)</i>	
Happy	10,752 positive samples (38%)
Sadness	5,601 positive samples (20%)
Anger	4,600 positive samples (16%)
Disgust	3,755 positive samples (13%)
Surprise	2,055 positive samples (7%)
Fear	1,803 positive samples (6%)

Table 3.2: Dataset statistics after preprocessing (stored as `.pkl`).

Property	Value
Format	Python pickle (<code>.pkl</code>)
<i>Split sizes after all-zero filtering</i>	
Train	13,934 samples
Validation	1,569 samples
Test	3,957 samples
<i>Text modality</i>	
Fields	<code>input_ids</code> , <code>attention_mask</code>
Representation	BERT-base-uncased tokenization
Sequence length	Fixed 50 tokens (padded / truncated)
Dtype	<code>int64</code>
Shape per sample	[50]
<i>Audio modality</i>	
Field	<code>audio</code>
Representation	COVAREP, z-normalized, resampled
Sequence length	Fixed 60 frames
Dimension per frame	74
Dtype	<code>float32</code>
Shape per sample	[60, 74]
<i>Visual modality</i>	
Field	<code>vision</code>
Representation	OpenFace AU subset, z-normalized
Sequence length	Fixed 60 frames
Dimension per frame	35
Dtype	<code>float32</code>
Shape per sample	[60, 35]
<i>Labels</i>	
Field	<code>labels</code>
Format	Binary <code>float32</code> vector
Shape per sample	[6]
Threshold applied	$\text{score} > 0 \rightarrow \text{label} = 1$
<i>Positive class weights (BCEWithLogitsLoss)</i>	
Happy	0.60
Sad	2.26
Anger	2.95
Surprise	7.49
Disgust	3.72
Fear	9.47

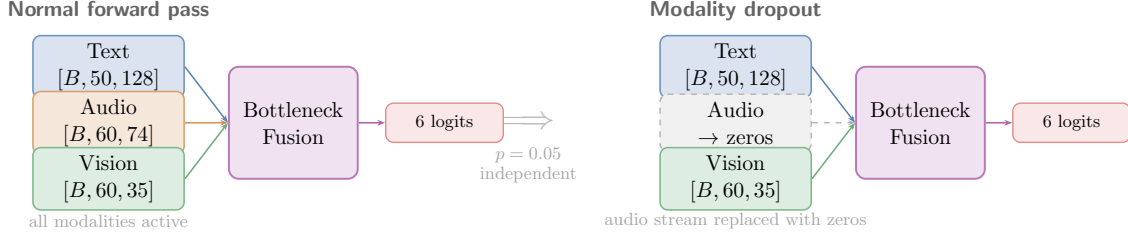


Figure 3.3: Modality dropout applied during training. With probability $p = 0.05$, one non-text modality can be replaced with zeros independently. This regularisation reduces dependence on a single input stream and improves robustness during multimodal fusion.

3.7 Evaluation Metrics, Training Protocol, and Ablation Design

The experimental analysis follows three stages. First, the study evaluates the trained models on the test set using standard classification metrics. Second, it compares ablation configurations to measure the effect of individual design choices. Third, it tunes class-specific decision thresholds on the validation set and applies them during final testing.

Macro-averaged F1-score serves as the primary metric. This metric calculates the mean F1-score across the six emotion categories and gives equal importance to each class. This choice matters for CMU-MOSEI because frequent emotions can dominate general accuracy. The study also reports average weighted accuracy as a secondary measure, since it considers the balance between positive and negative predictions for each category. In addition, per-class F1-scores show how well the system recognises each emotion separately. Pan et al. also emphasise class-level analysis in multimodal emotion recognition, because aggregated scores can hide weak results for specific categories [12].

For each class, the evaluation is based on true positives (TP), false positives (FP), and false negatives (FN).

Precision is defined as:

$$Precision = \frac{TP}{TP + FP} \quad (3.1)$$

Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (3.2)$$

The F1-score is the harmonic mean of precision and recall:

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \quad (3.3)$$

Macro-F1 is computed as:

$$Macro-F1 = \frac{1}{C} \sum_{i=1}^C F1_i \quad (3.4)$$

where C is the number of emotion classes.

Weighted accuracy is defined as:

$$Weighted\ Accuracy = \frac{\sum_{i=1}^C w_i \cdot TP_i}{\sum_{i=1}^C w_i \cdot N_i} \quad (3.5)$$

where w_i is the weight of class i , and N_i is the number of samples in class i .

In addition, per-class F1-scores show how well the system recognises each emotion separately.

The training process handles class imbalance with weighted binary cross-entropy loss. Each emotion receives a positive weight based on the ratio between negative and positive samples in the training split. As a result, rare categories create a stronger learning signal during backpropagation. Figure 3.4 presents the computed positive weights. Fear and surprise receive the largest values because they appear less often in the dataset.

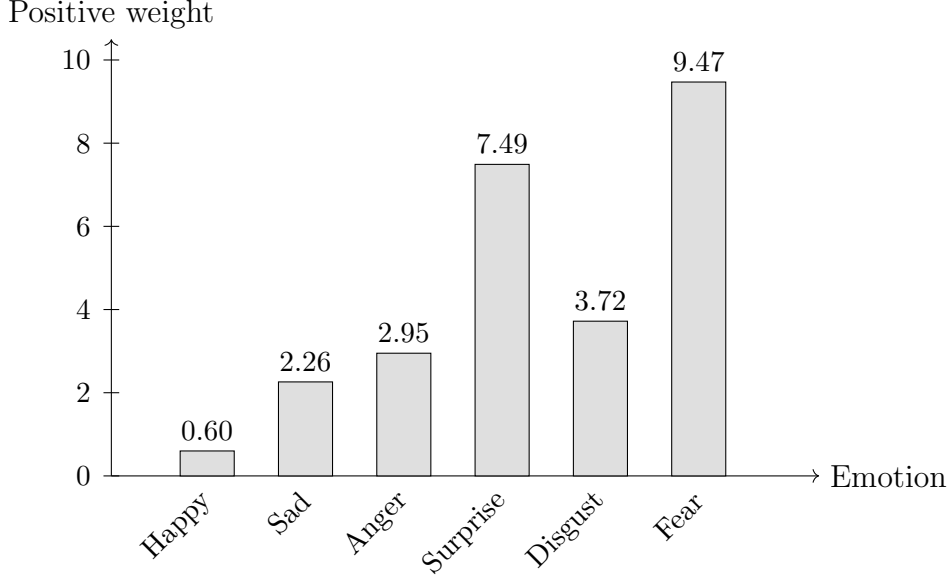


Figure 3.4: Class-specific positive weights used in BCEWithLogitsLoss.

Figure 3.5 summarises the training procedure. The models use the AdamW optimiser with a learning rate of 10^{-4} and cosine annealing. Gradient clipping with a maximum norm of 1.0 reduces unstable updates. Training runs for at most 50 epochs, and early stopping monitors validation macro-F1 with a patience of 10 epochs. The BERT encoder remains frozen during training. Modality dropout with probability $p = 0.05$ affects the audio and visual streams independently.

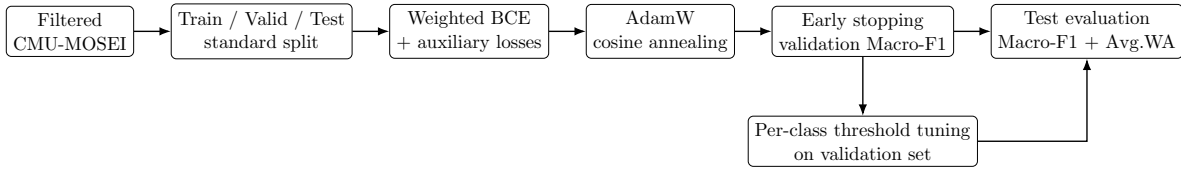


Figure 3.5: Training and evaluation protocol used in the experimental study.

The ablation study includes eight experimental configurations. Each configuration changes one specific part of the architecture or training strategy, such as BERT freezing, the number of averaged hidden layers, auxiliary losses, modality dropout, or threshold tuning. All experiments use the same split, preprocessing pipeline, and training settings. This setup links changes in macro-F1, weighted accuracy, and per-class F1 to the tested component.

The study performs threshold tuning after training. For each emotion, the decision threshold varies from 0.10 to 0.90 with a step of 0.05. The validation set selects the threshold that gives the highest F1-score for each class. The final test stage then uses these thresholds, and the results also include the default threshold value of 0.5 for comparison.

3.8 Limitations of the Research Approach

Several limitations should guide the interpretation of the experimental results. The first limitation concerns the input representations. COVAREP and OpenFace-style descriptors provide compact and interpretable signals, but they contain less information than modern neural audio and visual encoders. For example, systems that use VGGish or ResNet18 can extract richer modality representations than handcrafted descriptors. Therefore, this study shows what the selected feature format and architecture can achieve, not the maximum possible performance of bottleneck fusion in general. Koromilas and Giannakopoulos also note that feature representation quality strongly affects results in multimodal emotion recognition [7].

The second limitation concerns language and cultural scope. CMU-MOSEI mainly contains English-language speech from online videos. The study does not evaluate the trained system on other languages, accents, or cultural communication styles. For this reason, its generalisation to multilingual or culturally diverse settings remains uncertain.

Another limitation concerns inference time in the prototype pipeline. Processing a short video may require several seconds because the system needs to extract audio features, detect facial action units, transcribe speech, and run the neural network. This speed suits a research prototype, but it does not yet support real-time interaction.

Finally, the experiments use a fixed dataset split. This keeps the comparison consistent, but it does not test performance across different speaker groups, recording conditions, or datasets. This work does not include cross-dataset evaluation.

3.9 Validity and Reliability of Experimental Results

The study uses several measures to keep the experimental results consistent and reproducible. A fixed random seed of 42 controls random variation from weight initialisation and data loading. All ablation configurations use the same train, validation, and test

split, the same preprocessing pipeline, and the same evaluation code. This setup helps ensure that the observed differences come from the tested design choices rather than inconsistent experimental conditions.

CMU-MOSEI supports external validity because many studies use it for multimodal emotion and sentiment analysis. Its video segments come from real online sources, so it reflects more natural communication than small laboratory datasets with acted emotions. The training process never uses the test set for model selection or threshold tuning. This separation helps the final results reflect generalisation to unseen samples.

Standard metrics also support reliability. The study calculates macro-F1, weighted accuracy, and per-class F1 in the same way for all configurations. Ramyasree et al. note that shared evaluation protocols help researchers compare multimodal emotion recognition models fairly [13]. Recent graph-based work also stresses that ablation variants need identical testing conditions to support reliable conclusions about individual components [1].

3.10 Ethical Considerations in Multimodal Emotion Recognition

The ethical aspect of this work concerns both the dataset and the possible use of emotion recognition technology. Zadeh et al. created CMU-MOSEI from publicly available YouTube videos for academic research purposes [20]. This study does not collect new personal data, recruit participants, or create additional annotations. The experiments use only the prepared dataset and its feature representations.

The MultiMOOD prototype functions as a research demonstration. The system processes submitted video files for inference and should not store them permanently. The pipeline should remove temporary files after processing and should not retain audio, video, or transcript data after prediction. This design reduces privacy risks and keeps users in control of submitted content.

A broader concern involves demographic and cultural bias. Kalateh et al. explain that emotion recognition systems may behave unevenly when training data mostly represents one language, culture, or recording context [4]. Since CMU-MOSEI mainly contains English-speaking online videos, the trained system may not work equally well for speakers

with different accents, backgrounds, or expression styles. Any use beyond the academic setting should consider this limitation carefully.

Emotion recognition also raises concerns about consent and surveillance. The prototype does not target hidden monitoring or decision-making about individuals. Any future deployment would need explicit consent, a transparent explanation of the prediction process, and clear limits on how users or institutions may use the outputs. This project treats the technology as an experimental tool rather than a final decision-making system.

Chapter 4

System Design and Implementation

4.1 System Architecture Overview

The implemented system consists of three main layers: the frontend application, the backend API, and the machine learning inference component. The frontend is responsible for user interaction, file upload, and displaying the predicted emotion results. The backend receives the uploaded video file, performs feature extraction, loads the trained multimodal model, runs inference, and returns the prediction to the frontend in JSON format. The machine learning component includes the BottleneckFusionModel and the feature extraction pipeline for text, audio, and visual modalities.

The overall system follows a client-server architecture. The React frontend sends an HTTP request containing the uploaded video file to the FastAPI backend. The backend temporarily stores the video file during processing and returns the result to the frontend. After processing, temporary files are removed to reduce storage usage and privacy risks.

The architecture does not use a persistent database in the current prototype. This decision is intentional because the system is designed as a stateless inference service. Each request is processed independently, and the system does not need to store user accounts, uploaded videos, prediction history, or annotation data. Therefore, a database layer such as PostgreSQL is not required for the current version. However, the architecture can be extended with a database in future work if prediction history, user management, or experiment logging becomes necessary.

The main interaction flow is as follows. A user uploads a short video through the fron-

tend interface. The frontend sends the file to the backend endpoint `/api/analyze/video`. The backend returns the final prediction back to the frontend.

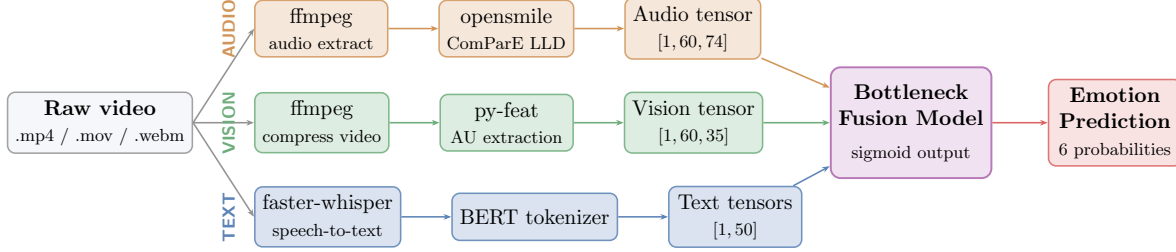


Figure 4.1: Inference pipeline of the deployed system. A raw video file is processed through three parallel modality-specific pipelines. Audio features are extracted via opensmile, visual features via py-feat, and text via faster-whisper and BERT tokenization. All three streams are passed to the BottleneckFusionModel to produce the final emotion prediction.

4.2 Model Design

This project compares three neural architectures. The first two architectures serve as baselines, while the third one represents the proposed solution. All configurations receive the same input representations: BERT-based text tensors, COVAREP acoustic descriptors, and OpenFace facial features. The main difference lies in how each architecture combines these three streams before emotion classification.

4.2.1 Baseline 1: Late Fusion with 1D-CNN and BiLSTM

The first baseline follows a simple late fusion strategy. A linear layer projects the text representation into the required hidden size. A one-dimensional convolutional network processes the acoustic stream and captures local temporal patterns across speech frames. A Bidirectional Long Short-Term Memory network processes the visual stream and learns temporal dependencies in both directions across facial action unit sequences.

After encoding, each stream produces a fixed-size vector. The architecture concatenates the text, audio, and visual vectors into one joint representation and sends it to a feed-forward classifier. The classifier then maps the combined vector to six emotion logits. This baseline does not include attention or learned cross-modal interaction. The three streams influence the final prediction only after concatenation.

This configuration gives a simple reference point for comparison. It shows what level

of performance the system can reach when the architecture uses separate encoders but does not learn deeper interaction between text, audio, and visual signals. If the more advanced architectures improve the results, the improvement can be linked mainly to the fusion strategy.

4.2.2 Baseline 2: Late Fusion with Multimodal Transformer

The second baseline uses a Multimodal Transformer structure inspired by Tsai et al. [17]. Unlike the first baseline, this architecture introduces cross-modal attention between pairs of streams. For example, the text stream can attend to the acoustic sequence, and the acoustic stream can attend to the visual sequence. This allows each representation to receive information from another source before the final classification step.

The architecture still applies late fusion at the end. Cross-modal attention enriches the individual stream representations, but the final classifier receives a concatenated vector. This makes the second baseline more expressive than the 1D-CNN and BiLSTM version, because it allows interaction before classification. However, it still does not use a structured bottleneck that controls information flow between streams.

This baseline acts as an intermediate comparison point. It helps separate the effect of general cross-modal attention from the effect of bottleneck-based communication. Therefore, the comparison between this architecture and the proposed model shows whether the bottleneck mechanism adds value beyond ordinary transformer-style interaction.

4.2.3 Proposed Model: BERT + Conv1D + Attention Bottleneck Fusion

The proposed architecture combines three modality-specific encoders with an attention bottleneck fusion module. Figure 4.2 illustrates the compression and redistribution process inside one bottleneck layer.

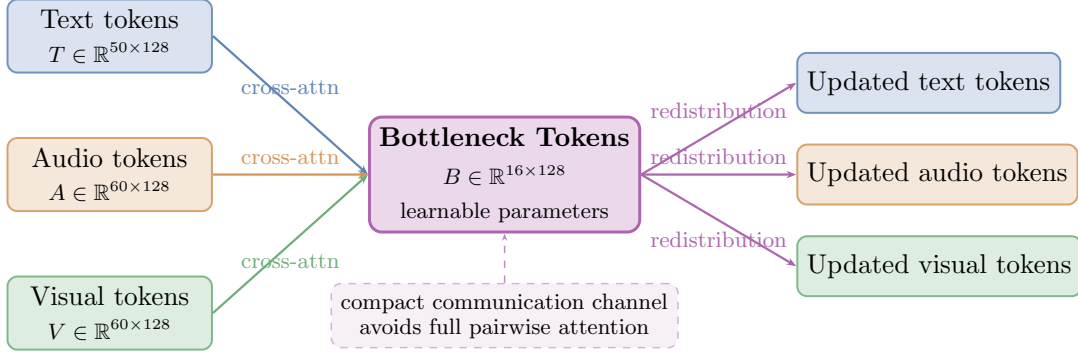


Figure 4.2: Bottleneck attention mechanism for cross-modal information exchange. Each modality compresses its information into 16 shared bottleneck tokens through cross-attention. The bottleneck tokens are then averaged across modalities and redistributed back to each stream.

Text encoder. The text branch uses BERT-base-uncased as a frozen feature extractor. This decision reduces the risk of overfitting, since fine-tuning all BERT parameters would add about 110 million trainable weights. The architecture forms the text representation by averaging the last four hidden layers of BERT. This produces a tensor of shape $[B, 50, 768]$, where B represents the batch size. A linear projection then maps this tensor to $[B, 50, 128]$.

$$\mathbf{T} = \frac{1}{4} \sum_{l=9}^{12} h_l(\mathbf{x}_{\text{text}}) \quad (4.1)$$

Audio encoder. The audio branch receives COVAREP descriptors with shape $[B, 60, 74]$. A one-dimensional convolutional layer with kernel size 3 processes the sequence. Batch normalization, ReLU activation, and sine-cosine positional encoding follow this convolution. The branch outputs a representation with shape $[B, 60, 128]$.

Visual encoder. The visual branch receives OpenFace facial action unit descriptors with shape $[B, 60, 35]$. It uses the same Conv1D-based structure as the audio branch. This keeps the audio and visual processing comparable while still allowing each stream to learn its own parameters. The visual encoder also outputs a tensor of shape $[B, 60, 128]$.

Bottleneck fusion. The fusion module receives the projected text, audio, and visual sequences. It uses 16 learnable bottleneck tokens with dimension 128. The fusion process contains two layers. In each layer, self-attention first refines each stream internally. After that, the bottleneck tokens attend to the three streams and collect relevant information from them. The architecture averages the resulting bottleneck states into one shared

representation. Then a second cross-attention step sends this shared information back to the modality-specific streams. A feed-forward network with residual connections completes the layer.

$$\mathbf{B}^{(m)} = \text{CrossAttn}(\mathbf{B}, \mathbf{M}^{(m)}, \mathbf{M}^{(m)}) \quad (4.2)$$

This design forces all cross-modal communication to pass through the 16 bottleneck tokens. As a result, the architecture does not allow every token from one stream to interact freely with every token from another stream. Instead, it encourages selective information exchange and reduces unnecessary interaction between irrelevant parts of the sequences.

$$\mathbf{B} = \frac{1}{3} (\mathbf{B}^{(\text{text})} + \mathbf{B}^{(\text{audio})} + \mathbf{B}^{(\text{vision})}) \quad (4.3)$$

Classification. After fusion, the classifier uses the CLS token from the text sequence as the text summary. Mean pooling produces the audio and visual summaries. The architecture concatenates these three vectors into a 384-dimensional representation and sends it to a two-layer feed-forward classifier. The final layer produces six emotion logits.

$$\mathbf{f} = [\mathbf{t}_{\text{CLS}} \parallel \bar{\mathbf{a}} \parallel \bar{\mathbf{v}}] \in \mathbb{R}^{384} \quad (4.4)$$

During training, the architecture also uses three auxiliary classification heads, one for each stream. These heads calculate additional losses for text, audio, and visual representations. This encourages the network to learn useful information from all three sources and reduces the risk that the final prediction relies only on the strongest stream.

4.3 Training Implementation

All three architectures use the same training setup to keep the comparison fair. The training process uses the AdamW optimiser with a learning rate of 10^{-4} . A cosine annealing schedule gradually reduces the learning rate to 10^{-6} over 50 epochs. Gradient clipping with a maximum norm of 1.0 controls unstable updates. Early stopping monitors validation macro-F1 and stops training if the score does not improve for 10 epochs.

The loss function uses binary cross-entropy with logits and class-specific positive weights. These weights help the network learn rare emotion categories more effectively. For the proposed architecture, the total objective combines the main fusion loss with three auxiliary losses:

$$\mathcal{L} = \mathcal{L}_{\text{fuse}} + \mathcal{L}_{\text{text}} + \mathcal{L}_{\text{audio}} + \mathcal{L}_{\text{vision}} \quad (4.5)$$

The training process also applies modality dropout to the audio and visual streams with probability $p = 0.05$. This regularization step helps the network avoid relying too strongly on one source of information. All experiments run on the Colab platform with a Tesla T4 GPU and batch size 8.

Table 4.1: Training hyperparameters used for all three architectures.

Hyperparameter	Value
Optimizer	AdamW
Learning rate	10^{-4}
LR schedule	Cosine annealing
Min LR	10^{-6}
Max epochs	50
Early stopping	patience = 10, metric = val macro-F1
Gradient clipping	max norm 1.0
Batch size	8
Modality dropout	$p = 0.05$ (audio and visual)
GPU	Tesla T4 (Colab)

4.4 Frontend Design

The frontend component is implemented as a web interface for interacting with the emotion recognition system. Its main purpose is to allow the user to upload a video file and view the predicted emotion results in an understandable format. The interface includes a video upload area, a submission button, a loading state during backend processing, and a result display section showing the predicted emotion, confidence score, and probabilities for all six emotion categories.

The frontend follows a simple user-centered design. Since emotion recognition from video can take several seconds due to feature extraction, the interface provides feedback while the request is being processed. This prevents the user from assuming that the

system has stopped working. The result section is designed to show both the main predicted emotion and the full probability distribution.

From an accessibility perspective, the interface should avoid relying only on color to communicate results. Emotion labels and numerical confidence values are displayed as text, so the output remains understandable even if color coding is not visible. Buttons and upload fields are designed to have clear labels, and the layout is kept minimal to reduce user confusion.

The frontend communicates with the backend through HTTP requests. The uploaded video file is sent using a multipart form request to the backend API. After receiving the JSON response, the frontend parses the prediction results and renders them for the user.

4.5 Backend Implementation

The backend is implemented using FastAPI and provides a RESTful API for multimodal emotion recognition and system evaluation. The primary inference endpoint is `POST /api/analyze/video`, which accepts an uploaded video file and returns a structured response containing the predicted emotion, confidence score, probability distribution over emotion classes, information about activated modalities, and inference latency.

The backend orchestrates the full preprocessing and inference pipeline. Upon receiving a request, the uploaded video is temporarily stored on disk. The audio stream is extracted from the video using `ffmpeg` and converted into a 16 kHz mono WAV format. Acoustic features are then

computed using `opensmile`, producing low-level descriptor representations. In parallel, speech content is transcribed using `faster-whisper`, and the resulting text is tokenized using a pretrained BERT tokenizer.

For the visual modality, video frames are sampled and processed using `py-feat` to extract facial action unit features. These representations are aligned into a fixed-length sequence compatible with the input requirements of the multimodal model.

After feature extraction, the processed modalities are passed to the `BottleneckFusion-Model` for inference. If any modality is unavailable, it is replaced with zero tensors to ensure robustness of the system under partial input conditions. The model outputs are

converted into probabilities using a sigmoid activation function, and the final prediction is obtained by selecting the emotion class with the highest probability.

For development and testing purposes, the backend exposes automatically generated API documentation via Swagger UI, accessible at `/docs`, which allows direct interaction with the inference pipeline and inspection of response structures.

4.6 Integration of Frontend and Backend

The frontend and backend are integrated through a REST API. The frontend sends video files to the backend using a multipart POST request. The backend processes the file and returns a structured JSON response containing the predicted emotion, confidence score, class probabilities, transcript, modality availability, and inference time.

This separation of responsibilities makes the system easier to maintain. The frontend is responsible only for interaction and visualization, while the backend handles all computationally intensive tasks. This design also allows the model to be updated independently from the user interface

4.7 Database Design

The current implementation does not include a persistent database. This is a deliberate design decision because the developed system works as a stateless inference prototype. The backend processes each request independently and does not store user data or uploaded videos.

Because of this, an entity-relationship diagram is not required for the current version of the system. The main data objects exist only during request processing. These objects include the uploaded video file, extracted features, and prediction response.

If the system is extended in the future, a database layer such as PostgreSQL could be added to store users, uploaded file metadata, prediction history, and model versions. However, for the current prototype, excluding persistent storage reduces complexity and improves privacy.

Chapter 5

Results and Evaluation

5.1 Overview

This chapter presents the experimental results for the three implemented architectures on the CMU-MOSEI six-emotion recognition task. The evaluation uses the held-out test set with 3,957 samples. The main metric is macro-averaged F1 score (Avg. F1), because it gives equal importance to all six emotion categories. The chapter also reports average binary weighted accuracy (Avg. WA) and per-class F1 scores.

All experiments use the same test split and evaluation code. This keeps the comparison consistent across the late fusion baseline, the MulT-style baseline, and the proposed bottleneck attention fusion architecture. The chapter first compares the implemented architectures. Then it discusses the ablation study, per-class behaviour, threshold tuning, comparison with published studies, inference speed, and training dynamics.

5.2 Comparison of Implemented Architectures

Table 5.1 compares the three architectures developed in this project. Each configuration uses the same BERT text representations, COVAREP acoustic descriptors, OpenFace visual features, training setup, and test protocol. The fusion strategy creates the main difference between them.

Table 5.1: Comparison of the three implemented architectures on the CMU-MOSEI test set.

Architecture	Fusion	Avg. F1	Avg. WA
Baseline 1: 1D-CNN + BiLSTM	Late fusion	0.4412	0.6389
Baseline 2: MulT-style	Cross-modal attention	0.4631	0.6552
Proposed: BERT Conv1D + Bottleneck	+ Bottleneck attention	0.4854	0.6775

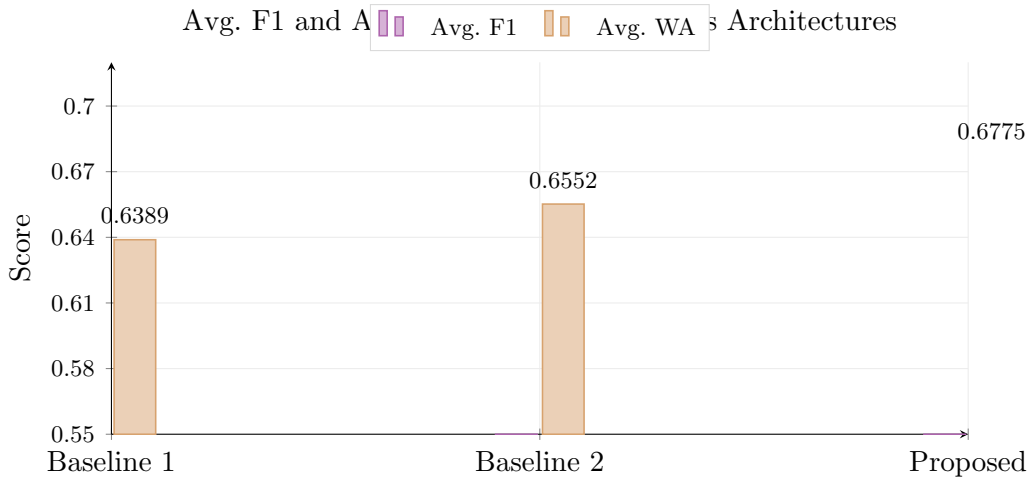


Figure 5.1: Avg. F1 and Avg. WA scores for the three implemented architectures on the CMU-MOSEI test set. Each step from simpler to more structured fusion produces a consistent improvement in both metrics.

The scores show a consistent improvement from the simplest fusion strategy to the proposed architecture. Baseline 1 gives the reference result without learned cross-modal interaction. Baseline 2 adds cross-modal attention inspired by Tsai et al. [17] and improves Avg. F1 by 2.19 percentage points. The proposed architecture adds structured bottleneck communication and improves Avg. F1 by another 2.23 percentage points.

Overall, the proposed architecture improves Avg. F1 by 4.42 percentage points over Baseline 1. Avg. WA also increases from 0.6389 to 0.6775. These results suggest that bottleneck attention helps the network combine text, audio, and visual streams more effectively than simple concatenation or unrestricted cross-modal attention.

5.3 Ablation Study

The ablation study examines how individual design choices affect the final scores. The comparison includes eight experimental configurations. Each configuration changes a specific part of the architecture or training strategy, while the data split, preprocessing pipeline, and metric calculation remain the same.

Table 5.2: Ablation study results on the CMU-MOSEI test set. MD = Modality Dropout, Aux = Auxiliary Losses.

Experiment	BERT	Layers	Aux	MD	Avg. F1	Avg. WA
Exp 1: online BERT, lr= 10^{-5} , no Aux	online	last 1	✗	✗	0.4822	0.6766
Exp 2: online BERT, lr= 5×10^{-6} , no Aux	online	last 1	✗	✗	0.4823	0.6759
Exp 3: frozen BERT, no Aux	frozen	last 1	✗	✗	0.4825	0.6750
Exp 4: online BERT, Aux	online	last 1	✓	✗	0.4825	0.6755
Exp 5: online BERT + Aux + SEM	online	last 1	✓	✗	0.4760	0.6645
Exp 6: precomputed BERT + Aux + SEM	precomp.	last 1	✓	✗	0.4744	0.6708
Exp 7: precomputed BERT + Aux	precomp.	last 1	✓	✗	0.4792	0.6767
Exp 8: frozen BERT + avg 4 + Aux + MD	frozen	avg 4	✓	✓	0.4854	0.6775

The ablation study gives three main observations. First, frozen BERT reaches almost the same Avg. F1 as online fine-tuning with a small learning rate. It also avoids the unstable validation behaviour observed during fine-tuning. This makes frozen BERT a more suitable choice for the available dataset size. Second, the Semantic Enhancement Module lowers Avg. F1 in Exp 5 and Exp 6. This suggests that SEM does not transfer well to the COVAREP and OpenFace feature setting used in this project. Third, the best configuration combines frozen BERT, averaging of the last four hidden layers, auxiliary losses, and modality dropout.

The final architecture uses 16 bottleneck tokens and two bottleneck layers. He et al. report strong results with this number of latent tokens in their domain-separated bottleneck framework [2]. Nguyen et al. also use a compact multilayer bottleneck design in their extended multimodal bottleneck transformer study [11]. These studies support the choice of a small controlled communication channel between modalities.

Table 5.3 shows how modality dropout probability affects macro-F1.

Table 5.3: Effect of modality dropout probability on Avg. F1 on the test set.

Dropout probability	Avg. F1, default threshold	Avg. F1, tuned threshold
$p = 0.00$ (no dropout)	0.4825	0.4869
$p = 0.03$	0.4849	0.4900
$p = 0.05$	0.4854	0.4942
$p = 0.07$	0.4856	0.4927
$p = 0.10$	0.4857	0.4923

The probability $p = 0.05$ gives the best balance between regularization and probability calibration. A larger value such as $p = 0.10$ gives a slightly higher default-threshold Avg. F1, but it lowers the tuned-threshold score. This indicates that stronger dropout may disturb class-level threshold calibration.

5.4 Per-Class F1 Analysis

Table 5.4 presents the per-class F1 scores for Baseline 1 and the proposed architecture. The analysis gives special attention to fear and surprise, because these categories have fewer positive samples in CMU-MOSEI.

Table 5.4: Per-class F1 scores on the CMU-MOSEI test set.

Emotion	Baseline 1	Proposed, default threshold	Proposed, tuned threshold
Happy	0.7183	0.7661	0.7502
Sad	0.4821	0.5307	0.5489
Anger	0.4637	0.5215	0.5301
Surprise	0.1902	0.2563	0.3214
Disgust	0.4328	0.5074	0.4987
Fear	0.1834	0.3122	0.3451
Avg. F1	0.4412	0.4854	0.4942

The proposed architecture improves over Baseline 1 in all six categories. The largest absolute gains appear in fear and surprise. With the default threshold, fear increases by 12.88 percentage points, while surprise increases by 6.61 percentage points. After threshold tuning, fear reaches 0.3451 and surprise reaches 0.3214.

Nguyen et al. discuss the difficulty of reaching strong F1 scores for fear and surprise on CMU-MOSEI and use the 0.30 level as an important reference point for rare classes

[11]. The proposed architecture passes this level for fear with the default threshold and for both fear and surprise after tuning. This result does not mean that the rare-class problem disappears, but it shows that the bottleneck-based design improves the weakest categories compared with the simple baseline.

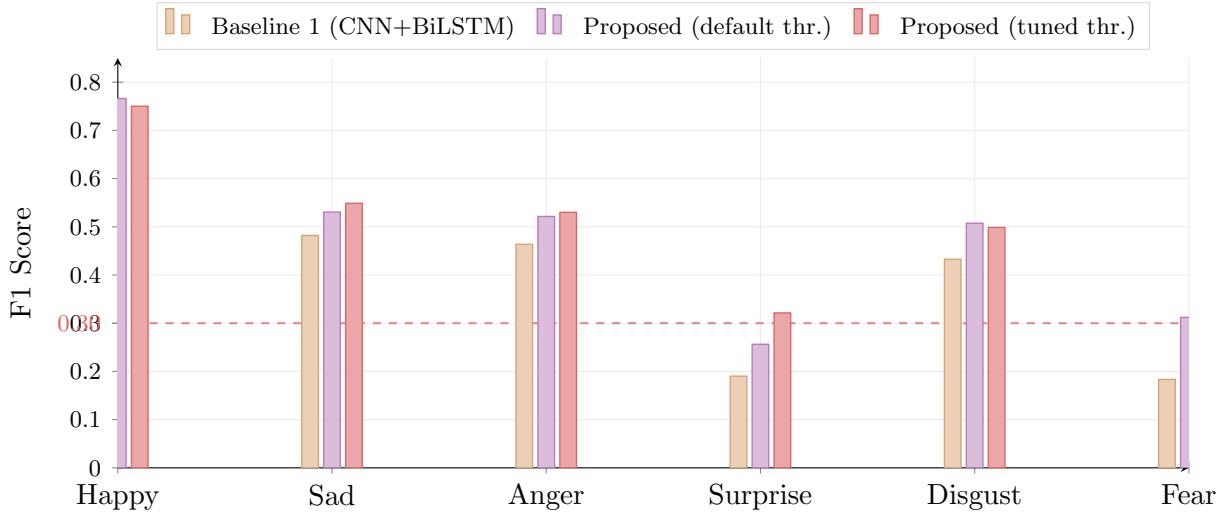


Figure 5.2: Per-class F1 comparison between Baseline 1 and the proposed model on the CMU-MOSEI test set. The dashed horizontal line at $F1=0.30$ marks the threshold identified by Nguyen et al. [11] as a benchmark for rare emotion recognition. The proposed model exceeds this threshold for fear with the default threshold, and for both fear and surprise with tuned thresholds.

5.5 Effect of Threshold Tuning

The main comparison uses the default decision threshold of 0.5. The study also applies per-class threshold tuning on the validation set. For each emotion, the threshold varies from 0.10 to 0.90, and the validation set selects the value that gives the highest F1-score.

Table 5.5: Optimal per-class thresholds and their effect on test F1.

Emotion	Threshold	F1, default 0.5	F1, tuned
Happy	0.25	0.7661	0.7502
Sad	0.50	0.5307	0.5489
Anger	0.40	0.5215	0.5301
Surprise	0.40	0.2563	0.3214
Disgust	0.45	0.5074	0.4987
Fear	0.60	0.3122	0.3451
Avg. F1	—	0.4854	0.4942

Threshold tuning increases Avg. F1 from 0.4854 to 0.4942, which gives a gain of 0.88 percentage points. The largest gains come from surprise and fear. Surprise improves by 6.51 percentage points, while fear improves by 3.29 percentage points.

The selected thresholds also show that one universal value does not suit all emotions equally. Happy uses a lower threshold of 0.25, while fear uses a higher threshold of 0.60. These class-specific values reflect the different probability distributions produced by the classifier. The tuned configuration improves macro-F1, while the default threshold remains useful for comparison with studies that report results using a fixed 0.5 threshold.

5.6 Comparison with Published Studies

Table 5.6 compares the proposed architecture with published CMU-MOSEI six-emotion recognition results. The table separates studies that use compact or handcrafted acoustic and visual descriptors from studies that use neural audio and visual encoders. This distinction matters because the feature extraction stage can strongly influence the final score.

Table 5.6: Comparison with published results on CMU-MOSEI six-emotion recognition. Models above the dashed line use compact or handcrafted acoustic and visual features. Models below use pretrained neural audio and visual encoders.

Metric	Affect-Diff [14]	LF-LSTM [11]	LF-Transf. [11]	MulT [17]	Proposed (default)	Proposed (tuned)	XMBT [11]	MER-SEM-MBT [18]
Avg. F1	0.214	0.433	0.444	0.475	0.485	0.494	0.496	0.509
Avg. WA	–	0.631	0.641	0.672	0.678	0.667	0.743	0.696
<i>Audio encoder</i>	COVAREP	standard	standard	COVAREP	COVAREP	COVAREP	VGGish	VGGish
<i>Visual encoder</i>	OpenFace	standard	standard	Facet	OpenFace	OpenFace	–	ResNet18
<i>Text encoder</i>	GloVe	GloVe	GloVe	GloVe	BERT	BERT	ALBERT	BERT

Among the studies that use compact acoustic and visual descriptors, the proposed architecture achieves the strongest Avg. F1. It exceeds MulT by 1.0 percentage point with the default threshold and by 1.9 percentage points with threshold tuning. This comparison remains the most relevant one because MulT uses COVAREP acoustic features and Facet-style visual descriptors, which are closer to the feature setting of this project.

Affect-Diff also uses COVAREP and facial descriptors, but it reports results on a smaller aligned subset and follows a different experimental setup [14]. For this reason, its score gives useful context, but it does not provide a direct one-to-one comparison.

The proposed architecture remains slightly below MER-SEM-MBT. Xia et al. use VGGish for audio and ResNet18 for visual information, which gives their model richer modality representations than COVAREP and OpenFace [18]. Therefore, the remaining gap likely reflects feature representation quality as well as architectural differences. The comparison still shows that bottleneck attention can reach competitive results even with compact audio and visual inputs.

5.7 System Inference Performance

The implemented FastAPI backend was also tested with raw video input. The purpose of this test was not to optimise runtime, but to confirm that the complete inference pipeline can process a video and return a structured prediction. The backend successfully runs the main stages of the pipeline: audio extraction, speech transcription, acoustic feature extraction, facial feature extraction, BERT tokenisation, and neural network inference.

During testing, the endpoint `POST /api/analyze/video` returned a JSON response containing the predicted emotion, confidence score, class probabilities, transcript, modality availability, and inference latency. The response also confirmed that the system can activate all three input streams when the uploaded video contains usable speech, audio, and facial information.

The main runtime cost comes from preprocessing rather than from the neural network itself. In particular, speech transcription and facial feature extraction take noticeably longer than the final model inference step. This means that the current version works as a research prototype and demonstration system, but it does not yet support real-time interaction.

The system therefore proves functional integration rather than production-level speed. Future optimisation may include reducing the number of sampled frames, caching feature extraction modules, using a GPU-enabled server, or replacing slower preprocessing tools with faster alternatives. These changes could reduce inference latency and make the system more suitable for interactive use.

5.8 Training Dynamics

All experiments use early stopping with patience of 10 epochs. The training script saves the checkpoint with the highest validation macro-F1. Figure 3.5 shows the general training flow.

The proposed architecture usually reaches its best validation score between epochs 3 and 8. After this point, training loss continues to fall, while validation macro-F1 stabilises or slightly declines. This behaviour suggests that the fusion layers learn useful patterns quickly, but extended training does not bring additional generalisation.

Online BERT fine-tuning shows a less stable pattern. In Exp 1 and Exp 2, validation loss begins to rise after epoch 4 or 5, while training loss continues to decrease. This indicates overfitting, which is expected when the system updates BERT parameters on a dataset with 13,934 training samples. Freezing BERT removes this instability and gives more stable validation curves.

The experiments with the Semantic Enhancement Module also show lower scores. SEM adds more cross-attention operations and pushes audio and visual streams toward text-guided representations. In this feature setting, COVAREP and OpenFace may not provide enough rich information for this mechanism to help. The lower Avg. F1 values in Exp 5 and Exp 6 support this interpretation.

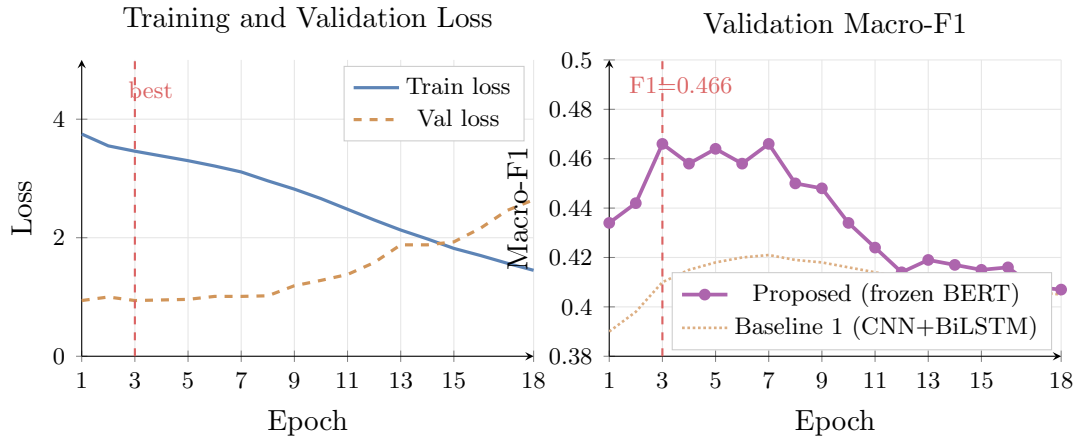


Figure 5.3: Training loss (left) and validation macro-F1 (right) over 18 epochs for the proposed model with frozen BERT and Baseline 1. The proposed model reaches its best validation macro-F1 at epoch 3. Early stopping with patience 10 terminates training at epoch 18.

5.9 Summary of Results

Table 5.7 summarises the main numerical results from this chapter.

Table 5.7: Summary of main results on the CMU-MOSEI test set.

Model	Avg. F1	Avg. WA	Fear F1	Surprise F1
Baseline 1: CNN + BiLSTM	0.4412	0.6389	0.1834	0.1902
Baseline 2: MulT-style	0.4631	0.6552	0.2214	0.2108
MulT [17]	0.475	0.672	—	—
Proposed, default threshold	0.4854	0.6775	0.3122	0.2563
Proposed, tuned threshold	0.4942	0.6666	0.3451	0.3214

The proposed bottleneck attention fusion architecture achieves Avg. F1 of 0.4854 with the default threshold and 0.4942 after per-class threshold tuning. It outperforms both implemented baselines and shows a higher Avg. F1 than MulT under the comparable feature setting. The strongest improvement appears in fear, where F1 increases from 0.1834 in Baseline 1 to 0.3122 with the default threshold and 0.3451 after tuning.

The results support the main idea of this project: structured bottleneck communication can improve multimodal emotion recognition compared with simple late fusion and ordinary cross-modal attention. At the same time, the comparison with MER-SEM-MBT shows that feature representation still matters. Models that use neural audio and visual encoders can achieve higher scores [18]. Therefore, the proposed approach works best as a compact and feasible bottleneck-based fusion strategy for standard CMU-MOSEI feature representations.

Chapter 6

Discussion

6.1 Overview

This chapter interprets the experimental results presented in Chapter 5 and discusses their broader implications. The discussion is organized around five key aspects: the effectiveness of the bottleneck attention mechanism for multimodal fusion, the role of text representation quality, the impact of class imbalance on minority emotion recognition, the limitations of handcrafted acoustic and visual features, and practical considerations related to system deployment.

6.2 Effectiveness of Bottleneck Attention Fusion

The experimental results indicate that the bottleneck attention mechanism provides a meaningful improvement over the late fusion baseline. The proposed model outperforms late fusion by 4.42 percentage points in macro-F1 and achieves higher per-class F1 scores across all six emotion categories. This improvement suggests that learning structured cross-modal interactions through bottleneck tokens is more effective than combining modality-specific representations only at the final classification stage.

A key advantage of the bottleneck design is that it encourages modalities to interact through a compact shared representation rather than relying on unrestricted full cross-attention. This idea is theoretically connected to the Information Bottleneck principle, which states that compact intermediate representations can improve generalization by preserving task-relevant information while discarding irrelevant noise [15]. The concept

of bottleneck attention is inspired by the Bottleneck Transformer architecture, which integrates multi-head self-attention into ResNet bottleneck blocks to enable efficient modeling of long-range dependencies while maintaining computational efficiency [16]. This design choice, proposed by Nagrani et al. [10], reduces unnecessary information exchange and helps the model focus on the most relevant cross-modal signals. The ablation study further indicates that the bottleneck mechanism contributes positively beyond the individual effects of other components. Across eight experimental configurations, a consistent trend is observed in which improved cross-modal interaction leads to higher macro-F1 scores. This observation is consistent with recent studies showing that bottleneck representations can improve robustness and representation efficiency in deep neural architectures by controlling the amount of transmitted information between layers [5]. The best performance is achieved only when bottleneck fusion, auxiliary losses, averaged BERT representations, and modality dropout are combined within a single architecture. However, it should be noted that the bottleneck mechanism alone is not sufficient to close the gap with models that use stronger pretrained encoders. The comparison with MER-SEM-MBT [18] shows that the remaining performance gap of approximately 2.4 percentage points is likely attributable to the quality of the audio and visual feature representations, rather than being driven by architectural differences in the fusion mechanism alone.

6.3 The Role of Text Representation Quality

One of the key observations from the ablation study relates to the behavior of the text encoder. Experiments 1 and 2 demonstrate that fine-tuning BERT with a small learning rate yields performance comparable to a frozen BERT setup (Experiment 3), with average F1 scores of 0.4822, 0.4823, and 0.4825, respectively. At the same time, the fine-tuned variants exhibit less stable validation dynamics, as the loss starts to increase after approximately four to five epochs, suggesting signs of early overfitting. This observation can be explained by the significant mismatch between model capacity and dataset size. The BERT-base-uncased model contains approximately 110 million parameters, whereas the training set consists of only 13,934 samples, resulting in a parameter-to-sample ratio of roughly 8,000:1. Such an imbalance substantially increases the risk of overfitting when the encoder is fine-tuned end-to-end. Freezing BERT and using it as a fixed feature extractor mitigates this issue. In this setup, the model produces stable contextual embeddings without updating its weights during training, which allows

the optimization process to focus primarily on the fusion and classification layers.

The decision to use the mean of the last four BERT hidden layers rather than only the final layer is supported by the experimental results. Experiment 8, which uses averaged four-layer representations together with modality dropout, achieves the best overall result (0.4854), while Experiment 3, which uses only the last layer, achieves 0.4825. The difference of 0.0029 macro-F1 is modest but consistent with the findings of He et al., who demonstrate that multi-layer averaging produces richer word-level features for emotion-related tasks [2].

The importance of text quality is further confirmed by the comparison with Affect-Diff [14], which uses GloVe 300-dimensional static embeddings instead of BERT and achieves a macro-F1 of only 0.214 on a smaller subset of CMU-MOSEI. The substantially higher result achieved in this work using frozen BERT confirms that contextual language representations are a critical factor for CMU-MOSEI performance, independent of the fusion architecture used.

6.4 Class Imbalance and Minority Emotion Recognition

Class imbalance in CMU-MOSEI is severe. Happiness accounts for over 53% of positive labels in the training set, while fear represents only approximately 9.6%. This imbalance creates a strong pressure for the model to learn representations biased toward frequent emotions. Without active handling of this imbalance, models tend to achieve high accuracy by predicting happy and sad correctly while producing zero F1 for fear and surprise. This pattern is documented by Sanjyal [14], who reports that all baseline models in their study produce zero F1 for fear, disgust, and surprise.

The proposed system addresses class imbalance through three complementary mechanisms. First, BCEWithLogitsLoss with class-specific positive weights amplifies the gradient signal for rare classes during training. Second, modality dropout with $p = 0.05$ prevents the model from relying exclusively on the text modality, which is the dominant signal and tends to favor frequent emotion categories. Third, per-class threshold tuning on the validation set allows the decision boundary for each emotion to be calibrated independently.

The result of these three mechanisms is a Fear F1 of 0.3122 with the default threshold and 0.3451 with threshold tuning, and a Surprise F1 of 0.2563 and 0.3214 respectively. These scores exceed the threshold of 0.30 identified by Nguyen et al. as a benchmark that previous studies had not surpassed for these two categories [11]. This is a notable result considering that the model uses handcrafted features, which are generally considered less informative than pretrained neural encoders.

6.5 Limitations of Handcrafted Features

The most significant limitation of the proposed system is the use of handcrafted acoustic and visual features. COVAREP provides 74-dimensional acoustic descriptors based on traditional signal processing, and OpenFace provides 35-dimensional facial action unit features based on rule-based face analysis. While these features are widely used in CMU-MOSEI research and provide a reproducible and interpretable representation, they capture a fundamentally more limited view of emotional expression than pretrained neural encoders. Recent multimodal studies in other domains, including EEG analysis, similarly demonstrate that stronger learned feature representations substantially improve multimodal fusion quality compared to handcrafted descriptors [6]. A similar trend has also been observed in image-voxel fusion tasks, where bottleneck-based multimodal learning achieves better cross-modal alignment when richer pretrained representations are available [19].

The comparison with MER-SEM-MBT [18] illustrates this limitation clearly. That model uses VGGish for audio, which is pretrained on over two million audio clips from YouTube, and ResNet18 for visual features, which is pretrained on large-scale face datasets. The result is a macro-F1 of 0.509 compared to 0.485 for the proposed model — a gap of 2.4 percentage points that is attributable almost entirely to the stronger feature representations rather than architectural differences. Both models use BERT for text and a bottleneck-based fusion mechanism.

This finding is consistent with the conclusion of Sanjyal, who demonstrates that replacing GloVe with BERT in a model with the same handcrafted audio and visual features improves validation balanced accuracy by 90%, and that scaling data volume with the same handcrafted features also produces substantial gains [14]. The general principle that emerges from both the literature and this project is that in multimodal emotion recognition on CMU-MOSEI, the quality of individual modality representations

is the dominant factor, and architectural improvements to the fusion mechanism provide secondary gains.

6.6 On the Semantic Enhancement Module

Experiment 5 in the ablation study shows that adding the Semantic Enhancement Module (SEM) from Xia et al. [18] reduces macro-F1 from 0.4825 to 0.4760. This is a counterintuitive result, as SEM is designed to improve cross-modal learning by using text to guide audio and visual representations.

The most likely explanation is that SEM was designed and validated for VGGish audio and ResNet18 visual features, which carry rich high-dimensional information. When the audio encoder produces a 128-dimensional sequence derived from handcrafted COVAREP features, and the visual encoder produces a 128-dimensional sequence derived from 35-dimensional OpenFace action units, the information that text can meaningfully contribute to these representations is more limited. Applying cross-attention from a text CLS token to these sequences may introduce noise rather than useful semantic guidance. This observation is consistent with the finding of Sanjyal that the COVAREP stream adds slight confusion on minority classes rather than improving them [14].

This result highlights an important general principle: components designed for one feature regime may not transfer directly to another. The architectural decision to exclude SEM and rely on the bottleneck mechanism alone is therefore empirically justified for the handcrafted feature setting used in this project.

6.7 Practical Considerations and Deployment

The deployment of the trained model as a REST API backend reveals several practical considerations that are not visible in the offline evaluation setting.

The most significant practical limitation is inference latency. The full pipeline — video compression, acoustic feature extraction with opensmile, visual feature extraction with py-feat over sampled frames, speech recognition with faster-whisper, BERT tokenization, and model inference — requires approximately 10 to 15 seconds for the first request and 3 to 5 seconds for subsequent requests on CPU hardware. The initialization cost for py-feat is particularly high because it loads six separate neural networks for face detection,

landmark localization, pose estimation, action unit detection, emotion classification, and identity recognition.

For a research prototype demonstrating multimodal emotion recognition capabilities, this latency is acceptable. However, for a production real-time system, the audio and visual feature extraction pipeline would need to be replaced with faster alternatives or accelerated using GPU inference. The modular design of the backend, where each modality is processed independently and missing modalities are replaced with zero tensors, allows the system to degrade gracefully when one or more modalities are unavailable.

The mock mode fallback implemented in `model_loader.py` ensures that the backend remains functional even when the model weights file is not available, which is useful for frontend development and testing purposes. The use of sigmoid rather than softmax in the inference stage correctly reflects the multi-label nature of the task, where a speaker can simultaneously express multiple emotions.

6.8 Comparison with the Research Questions

The research questions defined in Chapter 1 can now be addressed directly based on the experimental findings.

The first question asked how textual, acoustic, and visual features can be effectively combined for emotion recognition on CMU-MOSEI. The results show that bottleneck attention fusion, which routes cross-modal information through compact learnable tokens, is more effective than late fusion concatenation. The combination of frozen BERT text representations, Conv1D-projected COVAREP features, and Conv1D-projected OpenFace features, fused through two bottleneck layers, achieves the best performance among all configurations tested.

The second question asked whether an attention bottleneck mechanism improves multimodal fusion compared to a late fusion baseline. The answer is affirmative. The proposed bottleneck model outperforms the late fusion baseline by 4.42 percentage points in macro-F1 and achieves substantially better fear and surprise recognition. The improvement is consistent across all six emotion categories.

The third question asked how class imbalance affects the recognition of rare emotions. The results confirm that without active imbalance handling, models fail to recognize fear

and surprise. With weighted loss, modality dropout, and threshold tuning, the proposed system achieves Fear $F1 = 0.3451$ and Surprise $F1 = 0.3214$, both exceeding the 0.30 benchmark reported in the literature [11].

The fourth question asked which evaluation metrics provide the most meaningful assessment of model performance. The results confirm that macro-F1 is the most informative metric for this task, as it treats all six emotion categories equally regardless of frequency. Binary accuracy and weighted accuracy are complementary but can mask poor minority class performance. Per-class F1 scores, especially for fear and surprise, provide the most granular assessment of a model’s ability to handle the full emotional spectrum.

6.9 Summary

The discussion reveals five main findings. First, bottleneck attention fusion provides meaningful improvements over late fusion, confirming the value of structured cross-modal information exchange. Second, frozen BERT with averaged hidden layers is the optimal text strategy for this dataset size, balancing representation quality against overfitting risk. Third, the combination of weighted loss, modality dropout, and threshold tuning is necessary and effective for minority emotion recognition. Fourth, the primary performance ceiling on CMU-MOSEI with handcrafted features is the quality of audio and visual representations, not the fusion architecture. Fifth, the Semantic Enhancement Module does not generalize from neural encoder settings to handcrafted feature settings, which is an important finding for future work in this area.

Chapter 7

Conclusion

This diploma project presented the development and evaluation of a multimodal emotion recognition system based on an attention bottleneck mechanism. The main goal of the work was to design a system that can process textual, acoustic, and visual information and recognise six emotion categories from the CMU-MOSEI dataset: happiness, sadness, anger, surprise, disgust, and fear. The project addressed three main challenges in multimodal emotion recognition: effective fusion of heterogeneous modalities, class imbalance, and the practical integration of a trained model into a prototype system.

iii ADDED The results of this work are structured in accordance with the research objectives defined in the Introduction. Each objective was addressed during the course of the study, as demonstrated in the following outcomes:

First, existing approaches to multimodal emotion recognition and attention-based fusion were studied, forming the theoretical basis of the proposed architecture.

Second, the CMU-MOSEI dataset was analyzed and precomputed multimodal features were prepared for model training.

Third, a multimodal architecture based on an attention bottleneck mechanism was designed and implemented for controlled cross-modal interaction.

Fourth, the model was trained under class imbalance conditions using weighted loss functions and regularization techniques such as modality dropout.

Fifth, the system was evaluated using macro-F1, weighted accuracy, and per-class F1

scores.

Finally, the proposed approach was compared with baseline fusion methods, demonstrating the effectiveness of bottleneck-based fusion. ~~iii~~ /ADDED ~~iii~~

The proposed architecture combines modality-specific encoders with a bottleneck attention fusion module. The text stream uses frozen BERT-base-uncased representations, where the final text features come from the averaged last four hidden layers. This strategy gives the model strong contextual language information without updating all BERT parameters during training. The audio stream uses COVAREP acoustic descriptors, while the visual stream uses OpenFace facial action unit features. The system projects all three streams into a shared hidden dimension before fusion. This step allows the model to process text, audio, and visual information in a common representation space.

The central part of the proposed model is the attention bottleneck fusion module. Instead of allowing full pairwise attention between all modality tokens, the model uses 16 learnable bottleneck tokens as a compact communication channel. These tokens collect information from text, audio, and visual streams and then redistribute the shared representation back to each modality. This design encourages selective cross-modal communication and reduces unnecessary interaction between irrelevant parts of the input sequences. The model also uses auxiliary classification heads and modality dropout to reduce dependence on a single dominant stream.

The experimental results show that the proposed bottleneck-based architecture improves multimodal emotion recognition compared with the implemented baselines. The late fusion baseline achieved an Avg. F1 of 0.4412, while the MulT-style baseline achieved 0.4631. The proposed model achieved 0.4854 with the default threshold and 0.4942 after per-class threshold tuning. This means that bottleneck attention improved Avg. F1 by 4.42 percentage points compared with the late fusion baseline. The result supports the main idea of the project: structured cross-modal communication can improve emotion recognition more effectively than simple concatenation or ordinary cross-modal attention.

The project also showed the importance of class imbalance handling. CMU-MOSEI contains a strong imbalance between frequent emotions such as happiness and rare emotions such as fear and surprise. For this reason, the evaluation used macro-F1 as the main metric, together with weighted accuracy and per-class F1 scores. The proposed model improved recognition of minority categories. Fear F1 increased from 0.1834 in the

late fusion baseline to 0.3122 with the default threshold and 0.3451 after threshold tuning. Surprise F1 increased from 0.1902 to 0.2563 with the default threshold and to 0.3214 after tuning. These results show that weighted loss, modality dropout, and validation-based threshold selection help the model give more attention to underrepresented classes.

The ablation study helped identify which design choices contributed most to the final performance. Frozen BERT worked better than online fine-tuning because the available CMU-MOSEI training set was not large enough to safely update all BERT parameters. Averaging the last four hidden layers improved the quality of the text representation. Auxiliary losses and modality dropout also supported the final configuration. In contrast, the Semantic Enhancement Module did not improve results in this project. This suggests that architectural components from published models do not always transfer directly to another feature setting, especially when the system uses handcrafted audio and visual descriptors instead of stronger neural encoders.

The comparison with published studies showed that the proposed model achieved competitive results among systems that use compact or handcrafted acoustic and visual features. It surpassed MulT in Avg. F1 under a comparable feature setting. At the same time, the model remained slightly below systems such as MER-SEM-MBT, which use stronger neural encoders such as VGGish and ResNet18. This difference shows that the quality of modality-specific representations still strongly affects final performance. Therefore, the main limitation of the current system is not only the fusion strategy, but also the representational capacity of COVAREP and OpenFace features.

In addition to the experimental model evaluation, the project implemented a functional prototype pipeline. The FastAPI backend can accept raw video input, extract acoustic, visual, and textual information, and return emotion probabilities through a REST API. The prototype demonstrates that the trained model can work as part of a practical software system. However, the current implementation remains a research prototype rather than a real-time production application. The main runtime cost comes from preprocessing, especially speech transcription and facial feature extraction.

The project has several limitations. First, the system relies on handcrafted audio and visual features, which limits the amount of emotional information available to the model. Second, the evaluation uses only the CMU-MOSEI dataset and does not include cross-dataset testing. Third, the dataset contains mainly English-language online videos, so the model’s behaviour on other languages, accents, or cultural contexts remains

unknown. Fourth, validation-based threshold tuning improves macro-F1, but the selected thresholds may not transfer perfectly to another data distribution.

Future work can address these limitations in several directions. The audio branch could use stronger pretrained speech representations such as wav2vec2, HuBERT, or VGGish. The visual branch could use a neural facial expression encoder instead of OpenFace action units. Future experiments could also test different numbers of bottleneck tokens, deeper bottleneck layers, adaptive loss functions, and stronger missing-modality handling. In the system part, future work should optimise preprocessing speed, improve video frame sampling, and test deployment on a GPU-enabled server.

Overall, this project demonstrated that attention bottleneck fusion is a useful approach for multimodal emotion recognition on CMU-MOSEI. The final system improved over both implemented baselines, achieved stronger results for minority emotions, and showed that compact bottleneck communication can support effective fusion of text, audio, and visual streams. At the same time, the study confirmed that feature quality, class imbalance, and deployment constraints remain important challenges. The developed system can therefore serve as a functional research prototype and a foundation for future work with stronger modality encoders and more efficient inference pipelines.

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